

6 Further numerical methods	6.1	Numerical solution of first order and second order differential equations.	The approximations $\left(\frac{dy}{dx}\right)_n \approx \frac{(y_{n+1} - y_n)}{h}$ $\left(\frac{dy}{dx}\right)_n \approx \frac{(y_{n+1} - y_{n-1})}{2h}$ $\left(\frac{d^2y}{dx^2}\right)_n \approx \frac{(y_{n+1} - 2y_n + y_{n-1})}{h^2}$
	6.2	Simpson's rule.	